National University of Computer and Emerging Sciences, Lahore Campus

Course:	Linear Algebra	Course Code:	MT-1004
Program:	BS EE	Semester:	Spring 2023
Duration:	60 minutes	Total Marks:	40
Date:	27-02-2023	Weight	15%
Section:	All	Page(s):	2
Exam:	Midterm 1	Roll No:	22L-6265
Name:	Ali Anqa		

Instruction/Notes:

Attempt all questions. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem. Use of programmable calculators is not allowed. Exchange of stationary is strictly prohibited. Best of luck!

Question no. 1: (10 marks) (CLO #1)

Suppose that the augmented matrix of a system of linear equations has been partially reduced using elementary row operations to

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 2x-3 \\ 0 & 1 & 0 & y-2 \\ 0 & 0 & x & -y+1 \end{array}\right]$$

Find all values of x and y for which the given system has no solution, exactly one solution and infinitely many solutions.

1 & 2 $3 = 2x-3$
$1 = y-2$, $y=3$, $x = -y+1$

Question no. 2: (10 marks) (CLO #1)

Use inverse algorithm to find the inverse of given matrix. Also express the given matrix and its inverse as a product of elementary matrices.

$$\begin{bmatrix} 1 & 0 \\ 5 & -2 \end{bmatrix}$$

Question no. 3: (10 marks) (CLO #1)

Find the determinant of the given matrix by using row reduction method.

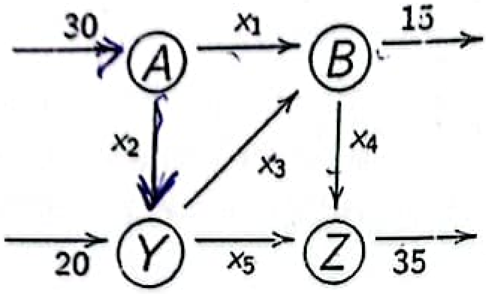
$$\begin{bmatrix} -1 & 2 & 4 & 6 \\ 0 & 0 & 1 & 7 \\ -1 & 2 & 4 & 14 \\ 0 & 2 & 4 & 6 \end{bmatrix}$$

Question no. 4: (10 marks) (CLO #1)

The flow of traffic through a network of telephone towers is shown in the following figure. Set up a linear system whose solution provides the unknown flow rates and also solve the system for the unknown flow rates by using Gauss elimination method.

$20 - x_5 - x_3 + x_2 = 0$

$x_5 + x_4 - 35 = 0$



$30 - x_1 - x_2 = 0$

$x_3 + x_1 - 15 = 0$

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	Course:	Linear Algebra	Course Code:	MT-1004
	Program:	BS EE	Semester:	Spring 2023
	Duration:	60 minutes	Total Marks:	40
	Date:	07-03-2023	Weight	15%
	Section:	All	Page(s):	2
	Exam:	Midterm-1 (Retake)	Roll No:	22L-6234.
Name:	HASEEB UR REHMAN			

Instruction/Notes:

Attempt all questions. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem. Use of programmable calculators is not allowed. Exchange of stationary is strictly prohibited. Best of luck!

Question no. 1: (10 marks) (CLO #1)

Suppose that the augmented matrix of a system of linear equations has been partially reduced using elementary row operations to

$$\left[\begin{array}{ccc|c} 1 & -1 & r-2 & 3 \\ 0 & 1 & 2s+1 & -1 \\ 0 & 0 & t-1 & s+3 \end{array}\right]$$

Find all values of s and t for which the given system has no solution, exactly one solution and infinitely many solutions.

Question no. 2: (10 marks) (CLO #1)

Find the inverse of the given matrix using inversion algorithm also express it as a product of the elementary matrices.

$$\begin{bmatrix} 3 & 2 \\ -7 & -5 \end{bmatrix}$$

Question no. 3: (10 marks) (CLO #1)

Find the determinant of the given matrix by using row reduction method.

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 1 & 2 & 3 \\ 4 & 5 & 4 & 3 \\ 2 & 2 & -4 & 5 \end{bmatrix}$$

Question no. 4: (10 marks) (CLO #1)

The flow of traffic through a network is shown in the following figure., Set up a linear system whose solution provides the unknown flow rates and also solve the system for the unknown flow rates by using Gauss elimination method.